

Louay Yahyaoui

Machine Learning Engineer

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Summary

Machine Learning Engineer with hands-on experience across the model lifecycle, from research and fine-tuning to production optimization for deployment. Experience includes computer vision, NLP and speech domains.

Work Experience

- **AI Engineer Intern (End-of-Studies Internship)** Mar 2025 - Sep 2025
EssilorLuxottica *Sophia-Antipolis, France*
 - Researched and implemented quantization techniques on real-time AI speech-enhancement models, achieving 4x size/activation compression with under 1% accuracy loss.
 - Built a reusable PyTorch and AIMET to ONNX export pipeline, adapted to the hardware team's requirements, enabling direct integration into the target SoC.
 - Tools:** PyTorch, ONNX, Qualcomm AIMET, OVHCloud, Git.
- **AI Research Intern** Jun 2024 - Aug 2024
MUST University *Tunis, Tunisia*
 - Reviewed research on AI applications in education and proposed an AI assistant to improve the university student experience, setting the technical direction for the project.
 - Built the assistant as an academic Q&A proof-of-concept by generating a GPT-4 synthetic dataset, fine-tuning Llama 3.1 (8B) with QLoRA, and integrating the model into a functional demo.
 - Tools:** PyTorch, Hugging Face Transformers, Weights & Biases.
- **Data Science Intern** Jun 2023 - Jul 2023
Hexastack *Tunis, Tunisia*
 - Developed an RNN language classifier for chatbot routing, reaching 99.5% accuracy on a small internal held-out set.
 - Updated the training and validation workflow, end-to-end, for an existing BERT intent detection and slot-filling pipeline, improving intent accuracy to 99.8% and NER accuracy to 97%.
 - Tools:** TensorFlow, Pandas, NumPy, Git.

Projects

- **RAG Pipeline over arXiv Papers** Aug 2026 - Present
Built a retrieval-augmented QA system with LangChain: chunks and embeds paper text and generates context-grounded answers via an open-source LLM (Qwen2.5-7B).
- **Deepfake Video Detector** May 2026 - Jul 2026
Reproduced the GenD paper by implementing a CLIP-based deepfake detector, fine-tuning only LayerNorm parameters (0.03% of weights), achieving 85.9% AUROC on DFDC.
- **Satellite Image Segmentation** Dec 2025 - Jan 2026
Trained a TensorFlow U-Net to segment complex geospatial structures on the DeepGlobe land cover dataset, attaining 81% IoU with strong spatial consistency.
- **Lightweight GPT-style Language Model** Nov 2025 - Jan 2026
Implemented a GPT-style language model on the TinyStories dataset (2.7M short stories) for end-to-end next-token prediction, yielding a validation perplexity of 3.9.
- **Compressed Sensing Image Reconstruction with Generative Models** Feb 2024 - May 2024
Implemented a variational autoencoder for compressed sensing, reconstructing MNIST digit images from 20 noisy measurements (vs. 784 total pixels), achieving high reconstruction fidelity.

Education

- **Engineering Diploma in Software Engineering** Sep 2020 - Oct 2025 (Graduated)
National Institute of Applied Science and Technology (INSAT)

Skills

- **ML/DL:** PyTorch, TensorFlow, Hugging Face, QLoRA, LangChain, Qdrant, prompt engineering
- **Software & Tooling:** Python, C++, TypeScript, SQL, Bash, Linux, Git, CI/CD, Docker
- **Cloud & Infra:** OVHCloud, Azure
- **Languages:** English (Fluent), French (Advanced), Arabic (Native), Spanish (Elementary)

Certifications

- **DeepLearning.AI:** Machine Learning, Deep Learning, NLP Specializations (Coursera 2024)